

Chapter 8

2

Discuss the roles of epidemiologists in the policy arena. How do their roles differ from the that of policy makers?

They can provide quantitative evidence for justifying needed policies. They can also speak on the effectiveness of a policy that was put in place. In short, they are meant to act as expert testimony and give necessary scientific evidence to support a policy.

This differs from the role of a policy maker who could look at current data and problems then create a policy to mitigate/prevent said problems. The policy makers are not responsible for creating a study to prove the importance of said policy but simply should analyze the current information and get expert opinion (epidemiologists) on the matter.

4

Describe the stages of the policy cycle. In which of these stages is the role of epidemiology most important?

Problem Definition

Here the policy makers define a problem they believe needs to be solved by putting a policy in place for. This means that the problem should be significant and have practical solutions.

Agenda setting

Set priorities and involve stakeholders to help determine what problems are most important to be dealt with and when.

Policy Establishment

The formal adoption of the policy/program/procedure.

Policy implementation

This stage is centered around actually enforcing and achieving the goals set out in the policy. This is done by government agencies.

Policy Assessment

Assessing / researching the effectiveness of the policy. Here this looks at if the policy had the effect it was designed to do.

While epidemiology plays an important role in most if not all of these steps, epidemiologists are most important in the policy assessment stage as they are responsible for collecting all the evidence for the assessments.

7

Provide an example of a public health law (or regulation) that was implemented in the past few years. Why was the law implemented? Do you think it was successful?

Due to COVID-19, there were policies that were implemented such as accelerated vaccine development, closures of schools, social distancing, and face masks. These were implemented to reduce the spread of the coronavirus and because there were large influx of cases around the world. In my opinion, the “quarantine” policy was not as effective in the US as it was the rest of the world because there was a lot of undermining the danger of the virus in the early stages and lack/inadequate contact tracing in the US which was implemented in other countries to a high degree of success.

10

Give two reasons why it is important for epidemiologic researchers to conform to high standards. What are three examples of ethical standards in epidemiology?

Why should epidemiologic researchers conform to high standards?

1. Promote the aims of research (knowledge, truth, avoidance of error)
2. Collaborative work (trust, accountability, respect)
3. Accountability to the public
4. Gain public support (if people can trust the quality of research then more funding and support is likely)
5. Moral and social values (human rights, social responsibility, health and safety)

What are examples of ethical standards in epidemiology?

1. Prohibitions against falsifying / fabricating data
2. Guidelines for authorship and copyrighting policies
3. Federal policies to prevent conflict of interest or human subject protections
4. Health and safety guidelines in place to prevent harm towards patients and subjects

Chapter 9

5

What are the most appropriate applications of mass screening and selective screening? Give one example of each.

Mass screening is when all members of a population group are screened regardless of risk status. Selective screening is when only members with high risk are screened. In general, screening should occur when there is a significant enough benefit to outweigh the cost of the screening. Most appropriate cases of mass screening is when either the cost of the screening is low so its easy to screen a relatively large population or when the population is small and there wants to be minimal risks. An example for these scenarios are temperature screenings at large events (test is cheap and non invasive so easy to do on population in mass) or thorough workplace screenings such as an exhaustive background check for say high security government agencies (background checks are a form of screening just not a health example). For selective screening, this is when there is an attempt to maximize the percent of positive tests. An example of this is getting screened for hereditary diseases if the disease runs in the family.

7

What is meant by overdiagnosis? Give an example of overdiagnosis that you have seen in the media.

Overiagnosis occurs when there is a positive diagnosis even if the disease is benign or minimal threat. This is a result of mass improvements in the technologies that can diagnosis disease. An example could be a benign tumor that is not life threatening or causing other sorts of health complications.

Problem

Calculate Sensitivity and Specificity

Test Result	Diseased	Not Diseased
Test Pos	350	1900
Test Neg	150	7600

$$\text{Sensitivity} = \frac{\#TruePos}{\#TruePos + \beta} = \frac{350}{350 + 150} = \frac{7}{10}$$

$$\text{Specificity} = \frac{\#TrueNeg}{\#TrueNeg + \alpha} = \frac{7600}{7600 + 1900} = \frac{4}{5}$$